Executive Summary:Predicting the outcome of Formula 1 races.

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GitHub: https://github.com/Merlin117/erdos_ds_f1

Introduction:

Formula 1 has long been considered the pinnacle of motorsport, and accordingly has a large and dedicated fanbase. Fans love Formula 1 in part because there is always the potential for an action-packed race. However, not all races live up to this potential, with some finishing in almost the exact same order the drivers started in, not unlike a Thanksgiving Day parade. This can be a particular problem for attracting new fans to the sport because if the first race they watch turns out to be “boring” on a surface level they may not give the sport a second chance.

The goal of our project was to determine if it is possible to predict how many overtakes are likely to occur in a race given the information available prior to its start, particularly from the free practice and qualifying sessions the drivers participate in on the two days before the race itself. We were drawn to this question because of the potential utility it would provide to a fan deciding whether to recommend that a friend new to Formula 1 watch a particular race.

Data Description and Model:

We used F1DB, an open source Formula 1 database that contains free practice data, qualifying data, race data, circuit information, and much more. We chose to use data from 1986 onwards due to significant race format differences prior to this cutoff.

In order to prepare the data for training, we had to perform some cleaning. In particular, the raw data included drivers who qualified but did not race, so we removed these entries and recalculated starting positions. Additionally we had to reconcile different qualifying and practice formats by selecting each driver’s best lap time from any session as their representative time.

Model:

The target variable for our model was the number of position changes in a race, defined by the difference between the starting and finishing orders. After iterating through multiple steps of feature and model selection, we found that the best-performing model was linear regression on the following features: median qualifying time gap, difference between pace ranking and starting position, circuit type, and interaction terms of these features.

The relevant features were calculated as follows: To obtain the median qualifying time gap, we calculated each driver’s gap in laptime to the next fastest driver then took the median for each race. This collapsed the lap times from up to 25 racers into a single number for each race. The difference between pace ranking and starting position was calculated by a similar method as the difference between starting and finishing position, i.e. by calculating on average how far each race participant’s starting position was from their position in the ranking of best lap times in practice and qualifying.

Model selection was performed using four-fold cross validation, then our final model was tested by training on the full training set before predicting on the testing set. The model was able to find a trend in position changes, but there remains a lot of unexplained variance as the r^2 score achieved was only 0.13.

Takeaways and Conclusion:

While our model is only mildly successful, we can still learn from it. Of the features we investigated, the most predictive feature was how different the starting grid was from the ranking of drivers based on their best lap time from practice and qualifying. The larger the difference, the more position changes were likely to occur during the race. This has a clear interpretation: When fast drivers face a mishap in qualifying or a penalty that drops them down the grid, their race revolves around recovering as many positions as possible, often leading to an exciting race. So, if a fan is waiting for a likely exciting race to watch with a non-fan friend, big upsets in qualifying are the main clue they should look for.

Now we should discuss why the model isn't more accurate. There are many possible reasons for this, but we believe that a major reason is that many events can occur during the race which are essentially random. As one example of these "random" events we examined the effect of DNF (Did Not Finish) results. When a DNF occurs in a race it leads to position changes which are not the result of the drivers' pace, introducing variance which cannot be accounted for since which drivers will DNF cannot be known ahead of time. To explore this, we fed our model a second set of data with all DNFs removed and positions adjusted to exclude the missing drivers. Our model performed better with this data, achieving an r^2 score of 0.20 on the test set, up from 0.13 for the base model.

This supports the idea that events which we can't expect to predict influence the outcome of the race and hinder efforts to accurately model overtakes. Aside from DNFs, there are many other such events that can occur, including: crashes or mechanical failures which don't cause a DNF but cause a driver to lose places, collisions with wildlife on the track, adverse weather events, safety cars, red flags due to trackside grass fires, etc. Indeed, one could make a strong argument that unexpected events like these that modify the expected outcome are a big part of the appeal of sports in general. As such, their presence in F1 shouldn't be surprising and arguably improves the sport, as even in a race that one would expect to be uneventful something very exciting might still happen.

Future Avenues for Exploration:

During the course of study we found there to be a negative linear relationship between DNFs and how much the finishing positions were changing from their starting grid. Thus, we identify some direction for future study:

- Coming up with historical features (historical average of DNFs per racetrack up to that race) or driver experience features (average number of races participated in) which may

exhibit a better relationship to our quantifier for ‘excitement.’ One approach to this may be to use machine learning to assist in finding such features.
- Exploring the predictability of DNFs for Formula 1 racing events, and then using the predicted values as a feature to predict our ‘excitement’ quantifier.

Furthermore, since condensing the data into useful features was a primary challenge, we expect that applying machine learning approaches to the problem would be a worthwhile endeavour as we may be able to uncover more predictive trends by utilizing more of the raw data rather than having to apply some intuition to create what we think will be good features.